

BUSINESS REQUIREMENTS DOCUMENT

E-Commerce Sales & Operations Performance Dashboard

Sample / Practice BRD — built from the cleaned Ecommerce Raw Dataset (5,000+ rows)

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Prepared By	Data & Insights Team
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Source Dataset	Ecommerce_Raw_Dataset_5000plus.xlsx (RawOrders, cleaned)

Document Control

Revision History

Version	Date	Author	Description of Change
0.1	—	Data & Insights Team	Initial draft based on cleaned dataset profiling
1.0	2026-09-09	Data & Insights Team	First complete draft circulated for stakeholder review

Approvals

Role	Name	Signature / Approval	Date
Business Sponsor	[TBD]	[Pending]	[TBD]
Product / Analytics Owner	[TBD]	[Pending]	[TBD]
BI Developer	[TBD]	[Pending]	[TBD]

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1. Introduction & Purpose

This Business Requirements Document (BRD) defines the business need, scope, data foundation, and functional requirements for an E-Commerce Sales & Operations Performance Dashboard. The dashboard will be built on top of the cleaned and transformed version of the Ecommerce_Raw_Dataset_5000plus.xlsx source file, which has already been de-duplicated, standardized, and split into a star schema (fact and dimension tables).

This document is intended to be used as a working template: sections marked [TBD] should be completed with real stakeholder names, dates, and sign-offs before development begins. All figures describing the dataset below (row counts, category lists, etc.) are drawn from a profile of the actual cleaned source file, so the requirements are grounded in what the data can support.

1.1 Background

The source data represents transactional order-line records for an online retailer: each row is one product line within a customer order, along with product, supplier, customer, payment, and fulfillment details. The raw export contained common data-quality issues (duplicate rows, inconsistent casing, mixed boolean formats, mixed date formats, inconsistent country/payment-method spellings, and some missing or invalid values), which have been resolved in the transformation stage that precedes this dashboard build.

1.2 Purpose of the Dashboard

The dashboard will give business stakeholders (commercial, operations, and customer service leadership) a single, self-service view of sales performance, order fulfillment health, product/category performance, supplier performance, and payment behavior — replacing manual spreadsheet reporting with a governed, refreshable BI solution.

2. Business Objectives

The dashboard should enable the business to answer the following questions on a recurring (weekly/monthly) basis:

- What is total revenue, order volume, and average order value, and how are they trending over time?
- Which product categories, products, and suppliers drive the most revenue and margin-relevant volume?
- How healthy is order fulfillment — what share of orders are delivered, cancelled, returned, or refunded, and is that changing?
- Which customers and customer segments are most valuable, and how concentrated is revenue among them?
- Which payment methods are customers using, and does payment method correlate with cancellations or returns?
- Where (by customer / order geography) is demand concentrated?

Success is defined as: business users can answer each question above without exporting data to Excel, in under 3 clicks / filter selections, on data no more than 24 hours old.

3. Project Scope

3.1 In Scope

- A cleaned, modeled dataset (1 fact table + dimension tables) as the single source of truth for the dashboard.

- An interactive BI dashboard (Power BI / Tableau / equivalent) covering Sales, Fulfillment & Returns, Product & Category, Supplier, Customer, and Payment perspectives.
- Standard global filters: date range, category, payment method, order status, and geography.
- Scheduled or on-demand data refresh from the cleaned dataset.
- Row-level export of underlying detail for any visual, for audit purposes.

3.2 Out of Scope (Phase 1)

- Predictive analytics / forecasting (e.g., demand forecasting, churn prediction) — candidate for Phase 2.
- Real-time (sub-hourly) streaming refresh — Phase 1 uses batch refresh.
- Profitability / margin analysis requiring cost data not present in the source file (e.g., unit cost, shipping cost).
- Write-back or transactional actions from the dashboard (e.g., issuing refunds).
- Mobile-native app; Phase 1 targets desktop/web and tablet-responsive layouts only.

4. Stakeholders & Roles

Stakeholder / Role	Interest in the Dashboard	RACI
Head of E-Commerce / Sponsor	Overall sales performance, category mix, growth trend	Accountable
Commercial / Sales Analysts	Revenue, AOV, product & category performance	Responsible
Operations / Fulfillment Lead	Delivery, cancellation, return, refund rates	Responsible
Customer Service Lead	Return/refund drivers by product & payment method	Consulted
Procurement / Supplier Manager	Supplier volume, reliability, geographic mix	Consulted
Finance	Revenue & discount trends for reconciliation	Consulted
BI Developer / Data Team	Data model, dashboard build, refresh reliability	Responsible
IT / Data Governance	Access control, PII handling, source system integrity	Informed

5. Data Source & Data Model

5.1 Source Dataset

Source file: Ecommerce_Raw_Dataset_5000plus.xlsx, sheet 'RawOrders'. Profile of the file at the time of writing:

- 5,304 raw rows (order-line grain) before de-duplication; ~104 exact duplicate rows removed during transformation.
- 4,232 distinct orders, 899 distinct customers, 150 distinct products, 40 distinct suppliers.
- 10 product categories: Apparel, Beauty, Books, Electronics, Footwear, Furniture, Grocery, Home & Kitchen, Sports, Toys.
- Payment methods standardized to: Credit Card, Debit Card, PayPal, Net Banking, UPI, Cash on Delivery.

- Fulfillment flags: isDelivered, isCancelled, isReturned, isRefunded (standardized to Boolean during transformation).

Known data-quality issues resolved in the transformation layer feeding this dashboard: duplicate rows, inconsistent name/category casing and whitespace, mixed boolean representations (True/False, Yes/No, 1/0), mixed date formats in OrderDate, inconsistent Payment Method / Country spellings, a small number of rows with Quantity = 0 or a negative Discount, and nulls in Description, CategoryName, SupplierAddress, CustomerPhone, CustomerEmail, Discount, FinalProductPrice, OrderAddress, and PaymentMethod. FinalProductPrice is recomputed as (ProductPrice × Quantity) – Discount wherever it was missing.

5.2 Target Star Schema

The dashboard will query a star schema rather than the flat raw export, so that measures aggregate correctly and slow-changing attributes (customer, product, supplier) don't duplicate fact rows.

Table	Grain / Contents	Key Fields
FactOrders	One row per order line	OrderLineID (PK), OrderID, ProductID (FK), SupplierID (FK), CustomerID (FK), PaymentID (FK), DateKey (FK), Quantity, ProductPrice, Discount, FinalProductPrice, isDelivered, isCancelled, isReturned, isRefunded
DimProduct	One row per product	ProductID (PK), ProductName, Description, CategoryName
DimSupplier	One row per supplier	SupplierID (PK), SupplierName, SupplierCountry
DimCustomer	One row per customer	CustomerID (PK), CustomerName, CustomerCountry, CustomerEmail, CustomerPhone
DimPayment	One row per payment method	PaymentID (PK), PaymentMethod (standardized)
DimDate	One row per calendar date	DateKey (PK), Date, Year, Quarter, Month, MonthName, WeekOfYear, DayOfWeek

Note: Country should be parsed out of the free-text Address fields (Customer/Supplier/Order Address) during transformation and stored as its own attribute on DimCustomer / DimSupplier / DimDate-adjacent order geography, rather than parsed at dashboard query time.

5.3 Refresh & Ownership

- Refresh cadence: daily batch refresh (Phase 1); revisit for near-real-time in a later phase if required.
- Data owner: BI Developer / Data Team, in coordination with the source system owner.
- Historical load: full history currently in the source file, loaded once; incremental loads thereafter.

6. Functional Requirements

6.1 Global Filters (apply across all pages)

- Date range (with quick presets: MTD, QTD, YTD, Last 12 Months)
- Category
- Payment Method
- Order Status (Delivered / Cancelled / Returned / Refunded / In Progress)
- Customer Geography (Country, parsed from address)

6.2 Core KPIs (header cards, visible on every page)

KPI	Definition
Total Revenue	SUM(FinalProductPrice) for non-cancelled order lines in the selected period
Total Orders	DISTINCTCOUNT(OrderID) in the selected period
Average Order Value (AOV)	Total Revenue ÷ Total Orders
Units Sold	SUM(Quantity) for non-cancelled order lines
Cancellation Rate	COUNT(isCancelled = True) ÷ Total Order Lines
Return Rate	COUNT(isReturned = True) ÷ COUNT(isDelivered = True)
Refund Rate	COUNT(isRefunded = True) ÷ Total Order Lines
Average Discount %	AVERAGE(Discount ÷ (ProductPrice × Quantity))

6.3 Dashboard Pages

Requirement REQ-1 — Executive Sales Overview

- Revenue and Orders trend over time (line chart, selectable Day/Week/Month).
- Revenue by Category (bar chart) and Category mix (share of total).
- Top 10 Products by Revenue (ranked bar / table).
- KPI header cards from section 6.2.

Requirement REQ-2 — Fulfillment & Returns

- Delivered / Cancelled / Returned / Refunded funnel or stacked bar, by month.
- Cancellation and Return rate trend over time, with a target/threshold reference line.
- Return rate by Category and by Payment Method (to spot problem areas).
- Table of top products by return count, with return rate %.

Requirement REQ-3 — Product & Category Performance

- Revenue, units sold, and average discount by Category and by Product, filterable and sortable.
- Price vs. units-sold scatter, to spot elasticity outliers.
- Category trend small multiples (revenue over time, one chart per category).

Requirement REQ-4 — Supplier Performance

- Revenue and order-line volume by Supplier.
- Cancellation/return rate by Supplier (proxy for supplier quality).
- Supplier geography map or table (country of supplier).

Requirement REQ-5 — Customer Insights

- New vs. returning customers by month (based on first order date).
- Revenue concentration: top 10% of customers' share of total revenue (Pareto view).
- Customer geography (country) revenue breakdown.
- Simple RFM-style segmentation table (Recency / Frequency / Monetary) if data volume supports it.

Requirement REQ-6 — Payment Behavior

- Revenue and order share by Payment Method.
- Cancellation/return rate by Payment Method (e.g., is Cash on Delivery associated with higher cancellations).
- Trend of payment-method mix over time.

6.4 Interactivity Requirements

- Cross-filtering: clicking a category, supplier, or payment method on any visual filters the rest of that page.
- Drill-through from any summary visual to an order-line level detail table.
- Export-to-CSV / Excel on every table visual, for offline analysis.
- Tooltips showing underlying values (not just the visual encoding) on hover.

7. Non-Functional Requirements

Category	Requirement
Performance	Pages load in under 5 seconds with the full dataset and default filters.
Availability	Dashboard available 99% of business hours; refresh failures alert the data team.
Security / Access	Row-level or role-based access if regional teams should only see their own geography (Phase 2 consideration); PII fields (CustomerEmail, CustomerPhone) masked or excluded from non-admin views.
Usability	Consistent color coding for order status across all pages (e.g., green = delivered, red = cancelled/returned/refunded).
Data Quality	Automated validation on refresh: no negative discounts, no Quantity = 0 rows reaching the model, FinalProductPrice always populated.
Maintainability	Data model and DAX/measure definitions documented so a new analyst can extend the dashboard without rework.
Compatibility	Renders correctly on desktop browser and tablet; optimized resolution 1920×1080.

8. Assumptions & Constraints**8.1 Assumptions**

- The transformation/cleaning step (duplicate removal, standardization, star-schema split) is completed and versioned before this dashboard is built.

- Country can be reliably parsed from the free-text address fields; where parsing fails, records are flagged "Unknown" rather than dropped.
- Order-level figures are approximated by aggregating order lines, since no separate order-header table exists in the source.
- Cost/margin data is not currently available, so profitability KPIs are excluded from Phase 1.

8.2 Constraints

- Single source file for Phase 1; no live connection to a production order-management system yet.
- Dataset covers a fixed historical window (as exported); no guarantee of ongoing daily feed until an integration is scoped.
- Tooling choice (Power BI / Tableau / Looker) to be confirmed with the BI team; this BRD is tool-agnostic.

9. Success Criteria / Acceptance Criteria

The dashboard will be considered ready for sign-off when:

1. All 6 pages in section 6.3 are built and match the agreed KPI definitions in section 6.2.
2. Global filters (section 6.1) apply consistently across every page without breaking a visual.
3. Totals on the dashboard reconcile to a manual SUM/COUNT check against the cleaned dataset within a 0.1% tolerance.
4. Refresh completes successfully end-to-end at least 5 consecutive times in UAT.
5. At least 3 representative business stakeholders (sponsor, ops lead, analyst) complete UAT and sign off.
6. PII fields are confirmed masked/restricted per the access model agreed in section 7.

10. Indicative Timeline

Phase	Activities	Duration (indicative)
1. Data Preparation	Finalize cleaning/transform rules, build star schema, validate row counts	Complete (prerequisite)
2. Requirements Sign-off	Review & approve this BRD with stakeholders	1 week
3. Design	Wireframes, KPI definitions, color/style guide	1 week
4. Build	Data model, measures, visuals for all 6 pages	2–3 weeks
5. UAT	Stakeholder testing against acceptance criteria (section 9)	1 week
6. Deployment & Handover	Publish, set refresh schedule, document, train users	1 week

11. Appendix A — Data Dictionary (Key Fields)

Field	Description
OrderLineID	Unique identifier for a single product line within an order
OrderID	Identifier shared by all lines belonging to the same customer order
ProductID / ProductName / CategoryName	Product identity and category classification
SupplierID / SupplierName	Supplier fulfilling the product line
CustomerID / CustomerName	Customer placing the order
OrderDate	Date the order was placed (standardized to a single date format)
ProductPrice	Unit list price of the product
Quantity	Units of the product ordered on this line
Discount	Discount amount applied to the line (currency, not %)
FinalProductPrice	$(\text{ProductPrice} \times \text{Quantity}) - \text{Discount}$
isDelivered / isCancelled / isReturned / isRefunded	Standardized Boolean fulfillment flags
PaymentMethod	Standardized payment channel used for the order

12. Appendix B — Open Questions for Stakeholders

7. Should returns/refunds be analyzed against the order date or the return/refund date (not currently in the source file)?
8. Is a cost or margin field available from another system that could be joined in for profitability KPIs?
9. What is the required data-retention / history window for trend analysis (e.g., rolling 24 months)?
10. Do regional teams require row-level security, or is a single shared view acceptable for Phase 1?